Research Ideas Discovery via Hierarchical Negative Correlation

Lyuzhou Chen, Xiangyu Wang, Taiyu Ban, Muhammad Usman, Shikang Liu, Derui Lyu, and Huanhuan Chen, *Senior Member, IEEE*

***Abstract*— A new research idea may be inspired by the connections of keywords. Link prediction discovers potential nonexisting links in an existing graph and has been applied in many applications. This article explores a method of discovering new research ideas based on link prediction, which predicts the possible connections of different keywords by analyzing the topological structure of the keyword graph. The patterns of links between keywords may be diversified due to different domains and different habits of authors. Therefore, it is often difficult for a single learner to extract diverse patterns of different research domains. To address this issue, groups of learners are organized with negative correlation to encourage the diversity of sublearners. Moreover, a hierarchical negative correlation mechanism is proposed to extract subgraph features in different order subgraphs, which improves the diversity by explicitly supervising the negative correlation on each layer of sublearners. Experiments are conducted to illustrate the effectiveness of the proposed model to discover new research ideas. Under the premise of ensuring the performance of the model, the proposed method consumes less time and computational cost compared with other ensemble methods.**

***Index Terms*— Ensemble learning, keyword graph, negative correlation learning (NCL), research idea discovery.**

I. Introduction

KEYWORDS in research publications briefly summarize the core content, such as research issues, background, and methods [1], [2]. The co-occurrence of two keywords in the same publication may indicate a strong connection.[1] In many cases, these strong correlations suggest that the research concepts represented by these keywords may have some research value. Therefore, exploring the possibility of keyword pairs in the same research publication may help to discover valuable association of research concepts, inspiring researchers to generate new research ideas [4].

Link prediction predicts the possibility of a new link between two nodes by analyzing the structure of the graph or network [5]. It has been effectively applied in different tasks, such as social network analysis and knowledge graph completion [6]–[8]. Due to the similarity of application scenarios, link prediction could effectively be used to discover new research ideas [9]. A topological graph could be constructed by connecting keywords, where the edges between keywords represent the relevance between two keywords. Furthermore, the possibility of the connections of different keywords could be predicted from the perspective of link prediction [10], as shown in Fig. 1.

However, some challenges may exist in the prediction of the keyword co-occurrence. For instance, the keyword graph may cover various research domains, whereas methods in different domains may lead to different patterns of keyword connections. Furthermore, researcher habits linked with keyword selection may also impact the patterns [11]–[15]. Such issues indicate that the patterns of new links between keywords may be complex and diverse. Therefore, it may be difficult for a single learner to extract these diverse patterns comprehensively.

Ensemble learning combines different sublearners to learn diversely, thereby improving the performance of the model [16]. Traditional ensemble learning methods often train multiple sublearners for better generalization [17], [18]. However, many current link prediction methods employ models such as neural networks to achieve automated graph feature extraction, which requires high computational resources and time. Therefore, it is challenging to utilize traditional ensemble learning models to construct a link prediction model.

Negative correlation learning (NCL) constructs negative correlations mechanism for different sublearners [19]. It encourages sublearners to learn different aspects of the data [20]. Introducing this mechanism in the research idea discovery may enhance the model's ability to extract diversified new link patterns, thereby effectively reducing the model's dependence on the number of learners. However, this method is inclined to use the negative correlation between the results of sublearners to ensure diversity. Due to the vanishing gradient that may occur in some models using the backpropagation algorithm, the influence of the negative correlation applied to the results

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The authors are with the School of Computer Science and Technology, University of Science and Technology of China, Hefei 230026, China (e-mail: hchen@ustc.edu.cn).

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[1]For example, "Hydroxychloroquine, azithromycin, and COVID-19" are three keywords of a paper [3]. "Hydroxychloroquine and azithromycin" represent medications, and "COVID-19" represents the issue of the research.

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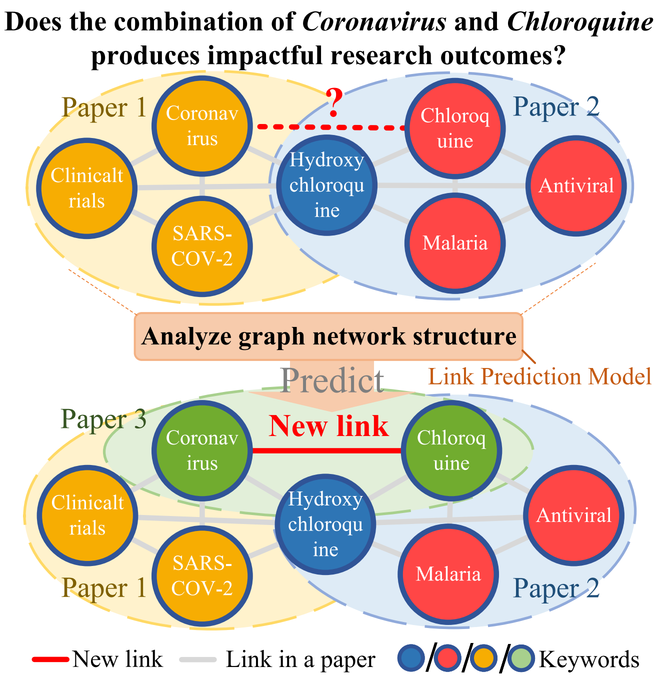

Fig. 1. Description of research ideas discovery.

tends to decrease layer by layer, which may make it difficult for the sublearners to have sufficient diversity [21]–[23].

To address these issues, this article constructs a negative correlation-based link prediction model to discover new research ideas. A hierarchical negative correlation (HNC) mechanism is designed, which considers the negative correlation between each order[2] subgraph features extracted by different learners and encourages ensemble of sublearners to extract diversified subgraph features. Through this mechanism, the weakness that the ensemble method consumes plenty of time and computational cost could be alleviated to some extent.

The contributions of this article are described as follows.

1) A method for discovering research ideas based on link prediction is proposed. It connects keywords of existing research publications in different domains through the analysis of the keyword graph, thereby inspiring new research ideas.
2) A negative correlation-based link prediction method is proposed to learn diversified graph features while reducing the computational resources and time consumption of the ensemble model in automatic graph feature extraction.
3) An HNC mechanism is proposed to encourage sublearners to extract different order subgraph features from different aspects, thereby ensuring the diversity of the ensemble model.

The remainder of this article is organized as follows. After introducing the related work of this article in Section II, the proposed model is illustrated in Section III. Experimental results and discussions are presented in Sections IV and V, respectively. Finally, Section VI concludes this article.

II. Related Work

Many traditional link prediction methods calculate the similarity of nodes to infer the possibility of links, which could mainly be divided into local- and global-based methods. Local-based methods mainly calculate the indices by employing information about common neighbors (CNs) and node degree, such as CN [24], Adamic–Adar (AA) [25], resource allocation (RA) [26], Jaccard [27], and preferential attachment (PA) [28]. Global-based methods, such as Katz [29], SimRank [30], and rooted PageRank (PR) [31], often employ the topological information of the entire network to calculate the similarity between nodes.

Many machine learning-based link prediction methods have been proposed in recent years, which automatically extract node features between nodes. Schlichtkrull *et al.* [32] used relational graph convolutional networks (R-GCNs) to predict links, which produces latent feature representations of nodes and utilizes a tensor factorization model to predict labeled edges. Kazemi and Poole [33] proposed SimplE for link prediction, an enhanced tensor factorization approach. This method learns two embeddings of each entity dependently and incorporates certain types of background knowledge into each embedding. Zhang and Chen [34] proposed SEAL, which maps the subgraph patterns to the status of link. This method learns embedding and gets a "heuristic" that suits the current network by extracting a local subgraph around each target link. In addition, many other methods, such as Weisfeiler–Lehman graph kernel (WLK) [35] and Weisfeiler–Lehman neural machine (WLNM) [36], are also proposed, which could be used for link prediction.

Ensemble learning combines multiple sublearners to improve the generalization of the model and obtains more accurate results. As an ensemble method, NCL is proposed by Liu and Yao [20]. This model trains the sublearners simultaneously and interactively through the correlation penalty terms to encourage diversity of the sublearners.

Many NCL-based methods have been proposed. To handle the possible overfitting problem in NCL, Chen and Yao [37] proposed the regularized negative correlation learning (RNCL), which incorporates an additional regularization term. Wang *et al.* [38] proposed the AdaBoost-NC, which introduces diversity by exploiting the ambiguity term in the decomposition. Chen *et al.* [39] proposed semisupervised NCL (SemiNCL), which considers the negative correlation terms for unlabeled data and extends the negative correlation mechanism to semisupervised learning tasks. Shi *et al.* [40] applied NCL into deep convolutional networks for crowd counting. The model could produce generalizable features and learn a pool of decorrelated regressors, which demonstrates the superiority over other methods.

III. Methods

A. Problem Definition

Suppose that the keyword graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_i, i \in \{1, 2, \ldots, N\}\}$ represents the set of

[2]For example, the kth order subgraph of a node composed of edges and nodes whose distance from the node does not exceed k.

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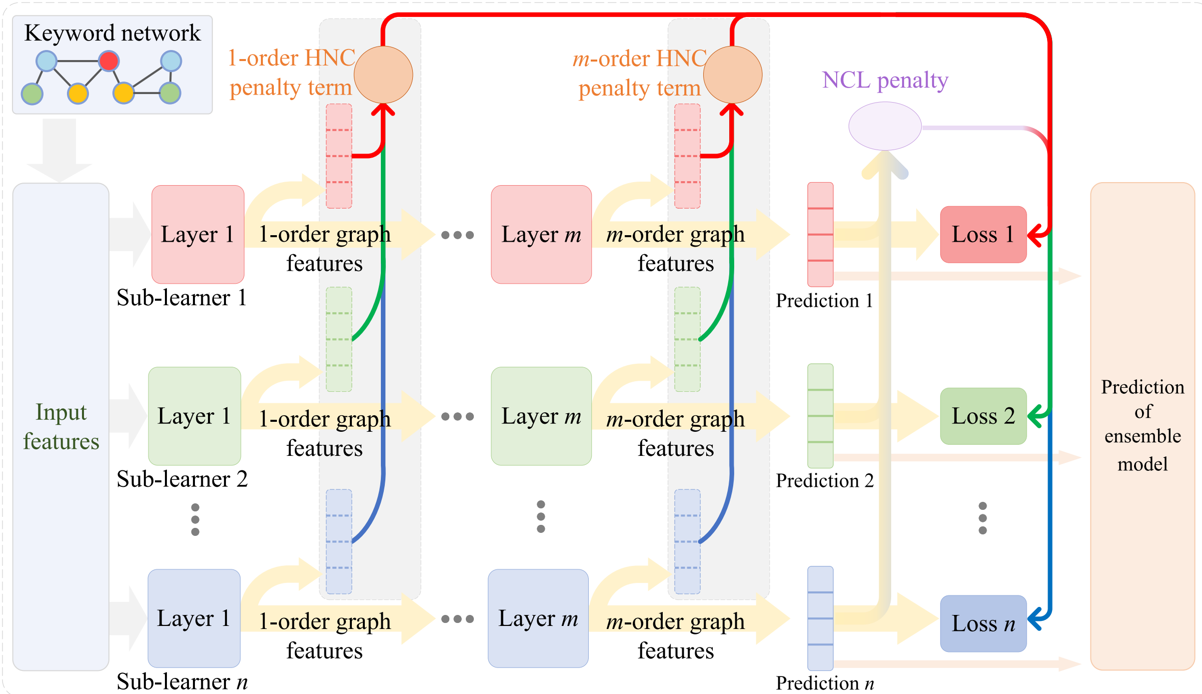

Fig. 2. Framework of the proposed model.

nodes (keywords) and $\mathcal{E} = \{e_{ij} \in \{1, 0\}\}$ represents the set of existing links. $e_{ij} = 1$ denotes a link between v_i and v_j, indicating that the corresponding two nodes appear in the same publication. $e_{ij} = 0$ denotes v_i and v_j that are not adjacent. The adjacency matrix $A \in \mathbb{R}^{N\times N}$ represents the topological structure of $\mathcal{G}$, where $A_{ij} = e_{ij}$. Given the adjacency matrix A and vertices $\mathcal{V}$, the goal of this task is to predict the possible new links appearing between two nodes, which could be expressed as

$$F = \mathcal{M}(A, \mathcal{X}) \tag{1}$$

where $\mathcal{M}(\cdot)$ represents the ensemble model to be constructed. $\mathcal{X} \in \mathbb{R}^{N\times d}$ is a matrix representing the input features of nodes $\mathcal{V}$, where d represents the number of features' dimensions.

B. Link Prediction Based on NCL

NCL constructs the ensemble learning model $\mathcal{M} = (\mathcal{M}_1, \mathcal{M}_2, \ldots, \mathcal{M}_n)$, which considers the negative correlation between sublearners $\mathcal{M}_i$ to improve the accuracy of ensemble. NCL is implemented by introducing a penalty term in the error function, which encourages sublearners to learn different aspects of given data. Given the training data $(A, \mathcal{X})$, the output of each sublearner is expressed as

$$f_i = \mathcal{M}_i(A, \mathcal{X}) \tag{2}$$

where $f_i \in \mathbb{R}^{|\mathcal{C}|}$ represents the predicted link status of the node pairs. The error function E_i for learner $\mathcal{M}_i$ is defined as

$$E_i = L(f_i, y) + \lambda p_i \tag{3}$$

where y represents the label of ground truth and $L(\cdot)$ represents the loss function of each sublearner. p_i is the penalty term, which indicates the penalty given to $\mathcal{M}_i$ from other sublearners, and λ ($\lambda \in [0, 1]$) is a scaling coefficient that adjusts the strength of the penalty. p_i is expressed as

$$p_i = (f_i - F)^T \sum_{j\neq i}(f_j - F) \tag{4}$$

where $F = (1/n)\sum_i f_i$ is the prediction of ensemble model. By introducing the penalty p_i, sublearners are expected to have diverse outputs under the premise of ensuring the accuracy of the model.

For sublearners, graph convolutional network (GCN) is utilized due to its good performance in extracting the graph structure. GCN extracts high-order features of subgraphs by aggregating features of the node and its neighbors' features in the convolutional layer. The features extracted by the lth layer of sublearner $\mathcal{M}_i$ are expressed as

$$O_i^{l+1} = \sigma\left(\tilde{\mathcal{D}}^{-\frac{1}{2}}\tilde{A}\tilde{\mathcal{D}}^{-\frac{1}{2}}O_i^l\mathcal{W}_i^{l+1}\right) \tag{5}$$

where $\tilde{A} = A + \mathcal{I}_N$ represents the adjacency matrix of graph $\mathcal{G}$ with added self-connections. $\tilde{\mathcal{D}}$ is a diagonal matrix where $\tilde{\mathcal{D}}_{jj} = \sum_k \tilde{A}_{jk}$. $\sigma(\cdot)$ is an activation function. l represents the lth layer of the sublearner. $\mathcal{W}_i^l$ is the parameter of lth layer. O_i^l denotes the features of l-order subgraphs. The negative correlation method using GCN as a sublearner is called NCL-GCN in the following.

C. HNC Mechanism

To encourage the diverse extraction of sublearners, an HNC mechanism is designed. It constructs negative correlation restrictions between features O_i^l of different-order subgraphs, It constructs negative correlation restrictions between features O_i^l of different-order subgraph, thereby promoting different sub-learners to extract the hierarchical features from different aspects, as depicted in Fig. 2.

The loss function $\tilde{E}_i$ of the sublearner M_i based on the HNC mechanism is defined as

$$\tilde{E}_i = E_i + E_i^H \tag{6}$$

where E_i is the loss function of NCL shown in (3). E_i^H is the HNC penalty term, which constructs negative correlations between different order graph features.

E_i^H acts on the subgraph, using the set $H_{ie} \subset \{1, 2, \ldots, m-1\}$ to indicate the order of the subgraph applied to the HNC mechanism, where m is the number of convolutional layers of GCN. For example, when $1 \in H_{ie}$, the HNC mechanism explicitly encourages the learner to extract the first-order subgraph features of the node with diversity. The expression of E_i^H is

$$E_i^H = \sum_{l \in H_{ie}} \mu^l C_i^l\left(O_i^l; O_j^l, j \neq i\right) + \sum_{l \in H_{ie}} S\left(O_i^l, y\right). \tag{7}$$

$C_i^l(O_i^l; O_j^l, j \neq i)$ is the separation term applied in the lth layer of GCN,[3] and μ is its weight. $S(O_i^l, y)$ is the restriction term that keeps the sublearner accurate. The separation term C_i^l represents the negative correlation between the subgraph features, which emphasizes the diversity. For $\mu^l (\mu^l > 0)$, μ^l controls the balance of diversity and accuracy. For example, when μ^l is large, the model tends to learn more diversely under the control of the HNC mechanism, while when μ^l is small, the model pays more attention to the accuracy of each sublearner. In particular, μ^l may also be set to a negative number, which minimizes the difference of outputs of different layers, thus making sublearners tend to be the same.

The restriction term $S(O_i^l, y)$ suppresses the learning direction of weak classification ability while learning different aspects of the data. The expression of $S(O_i, y)$ in (7) is

$$S(O_i, y) = L^H\left(h_i^l; y\right) \tag{8}$$

where h_i^l is the prediction result using the l-order subgraph features O_i^l. g_i^l is utilized to represent the corresponding prediction process, which could be thought of as a classifier, i.e., $h_i^l = g_i^l(O_i^l)$. L^H is the loss function. Note that h_i^l is only used to assist training, not the final prediction results of $\mathcal{M}$.

C_i^l in (7) is defined as

$$C_i^l\left(O_i^l; O_j^l, j \neq i\right) = \left(h_i^l - H^l\right)^T \sum_{j \neq i}\left(h_j^l - H^l\right) \tag{9}$$

where $H^l = \sum_i h_i^l$ is the average of the results based on O_i^l of each sublearner. The separation term C_i^l considers H^l and the result of other sublearners h_j^l $(j \neq i)$. Therefore, the sublearner may adjust the update direction and range of the parameter W_i^l according to the features learned by other learners. In this way, the diversity of subgraph features could be controlled. In the training process of the sublearner M_i, H^l and h_j^l are regarded as a constant. The effect of C_i^l is discussed in Section III-D.

The following content discusses the impact of separation term C_i^l on the model parameters W_i^l to show its characteristic in detail. The gradient in the backward propagation process is described as

$$\frac{\partial C_i^l}{\partial W_i^l} = 2\Phi^{AO^{l-1}} \frac{\partial f^l}{\partial O^l}\left(H^l - h^l\right) \tag{10}$$

where $\Phi^{AO^{l-1}}$ is used to represent the derivative of the subgraph feature O^l with respect to the parameter W_i^l.

In addition to being controlled by C_i^l, W_i^l is also affected by the separation term applied to other layers. Generally, the HNC separation term applied in the $(l+R)$ layer $(R \in \{1, 2, \ldots, m-1-l\})$ can be expressed as

$$\frac{\partial C_i^{l+R}}{\partial W_i^l} = 2\Phi^{AO^{l-1}} \Pi_{r=l+1}^{R+l} \Theta^{AW_i^r} \frac{\partial f^{l+R}}{\partial O^{l+R}}\left(H^{l+R} - h^{l+R}\right) \tag{11}$$

where Θ^{AW^r} represents the derivative of O^r with respect to O^{r-1}.

Equations (10) and (11) could demonstrate that the HNC separation terms of different layers have different impact on the features of a certain layer. Take the effect of C_i^{l+1} and C_i^l on the lth layer as examples. The first factor of $((\partial C_i^{l+1})/(\partial W_i^l))$ and $((\partial C_i^l)/(\partial W_i^l))$ is the same according to (10) and (11). The last factor of $((\partial C_i^{l+1})/(\partial W_i^l))$ and $((\partial C_i^l)/(\partial W_i^l))$ represents the negative correlation of the features of the corresponding layer. For the middle factor, it could be derived that the middle factor of $((\partial C_i^l)/(\partial W_i^l))$ is larger according to the nature of adjacency matrix A and the initialization method of parameter W_i^{l+1}.

Therefore, the lth separation term has a larger weight in training compared with $(l+1)$th. In other words, the influence degree of the separation term C_i^l at the lth level is greater than that of C_i^{l+1}. Similarly, it could be inferred that the influence of the separation term C_i^l on the l-order subgraph features gradually decreases as l decreases. Therefore, C_i^l is the main factor that affects the diversity of layer l.

D. Analysis for the Characteristic of HNC Mechanism

HNC mechanism applies negative correlation into multi-dimensional subgraph feature by separation term C_i^l. In the following, this article discusses the effectiveness of C_i^l due to its important nature in the HNC mechanism.

On the one hand, the separation term makes sublearners more diverse. The derivative of C_i^l with respect to h_i^l is[4]

$$\frac{\partial C_i}{\partial h_i} = 2(H - h_i). \tag{12}$$

Here, the component of h_i increases in the training process if it is larger than the average H and decreases if is lower. These characteristics make the prediction of sublearners different from the ensemble, showing the ability to encourage the diversity of sublearners.

On the other hand, the separation term obtains a balance between accuracy and diversity. The derivative of C_i to the parameters of the previous layer in the backpropagation process is

$$\frac{\partial C_i}{\partial t_j} = -\frac{1}{n|\mathcal{C}|} \sum_{k=1}^{|\mathcal{C}|}\left(h_i^k - H^k\right) \frac{\partial h_i^k}{\partial t_j} \tag{13}$$

[3]In the following, $C_i^l(O_i^l; O_j^l, j \neq i)$ is abbreviated as C_i^l.

[4]The superscript l is omitted for the convenience of description.

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where t_j is a parameter of the previous layer. $\mathcal{C}$ represents the set of possible labels and $|\mathcal{C}|$ represents its size. h_i^k and H^k represent the kth component of h_i and H. $h_i^l = g_i^l(O_i^l)$ and the last layer of g_i^l uses the softmax function in the proposed method.

According to the characteristics of the softmax function, the expression of $((\partial h_i^k)/(\partial t_j))$ is

$$\frac{\partial h_i^k}{\partial t_j} = \begin{cases} -h_i^k h_i^j, & j \neq k \\ h_i^k\left(1 - h_i^j\right), & j = k \end{cases} \tag{14}$$

when sublearner has enough diversity, i.e., the gap between h_i^k and H^k is large, h_i^k tends to be small or large. In this case, the effect of C_i decreases, which could be divided into several situations: 1) if h_i^k is small, the corresponding $((\partial h_i^k)/\partial t_j)$ is small, leading $(\partial C_i/\partial t_j)$ to be small; 2) if h_i^k is close to 1 and $j = k$, $((\partial h_i^k)/\partial t_j)$ is small; and 3) if h_i^k is close to 1 and $j \neq k$, it could be deduced that h_i^j is small since $\sum_k h_i^k = 1$. Therefore, $((\partial h_i^k)/\partial t_j)$ is small.

In summary, when the ensemble has enough diversity, the effect of the separation term C_i on the sublearners is reduced, which reflects the ability of the proposed method to achieve a balanced diversity and accuracy of sublearners.

IV. Experimental Results

The extensive experiment is conducted to evaluate the proposed method. AUC and average precision (AP) are employed as the evaluation metric. The influence of the four main parameters on the proposed model is analyzed according to the experimental results. Moreover, this article conducts the proposed method with different training methods to demonstrate the robustness of HNC. Three datasets are utilized, including COVID-19 Open Research, Computer Science (CS), and Protein–Protein Interaction (PPI) datasets.[5]

A. Datasets

Dataset (A): COVID-19 Open Research Dataset[6] is used for the experiments. It contains research publications on COVID-19 and related historical coronavirus. The theme of the publications includes various domains, such as medicine, virology, materials science, and social science.

The experiments utilize partial research publications in the dataset by collecting 1583 keywords as nodes of the graph. When two keywords co-occur in the same research publication, there is a link between the two corresponding nodes. Note that the constructed keyword graph is unweighted for links and contains 1583 nodes and 6859 links in total.

Dataset (B): This article collected publications from 36 CS journals and conferences and constructed the CS graph. The research domains of these publications include natural language processing (NLP), computer vision (CV), and social network analysis.

Partial research publications included in the dataset are utilized for experiments. The CS graph consists of 4153 nodes for corresponding CS keywords. The nodes that co-occur in the same publication are linked and the CS graph contains 27 006 links in total.

Dataset (C): PPI network is a network about whether there will be interactions between proteins, in which each node is a protein, and links between nodes indicate that the corresponding proteins are identified to have biologically significant interactions. A characteristic of PPI is that proteins with similar local graph structures are less likely to interact, which may contradict the assumptions of heuristics method. A subgraph of PPI is utilized in the experiment, which contains 4502 kinds of proteins and 7867 links in total.

B. Experimental Setup

In this work, random walk [41] is utilized to construct vector representations of the nodes in the graph as the input feature of sublearners, whose dimensionality is 64. Each graph comvolutional layer of sublearners extracts local subgraph feature and the dimensionality of the feature is 256. The module of graph convolution outputs the 256-D feature for each node, and then, a multilayer perceptron (MLP) is utilized to predict the occurrence of link with the concatenation of representations of two corresponding nodes as input. The proposed method uses different numbers of sublearners according to the specific settings of different experiments and simultaneously trains multiple sublearners. For the training process, the output of each graph convolution layer is saved to calculate the HNC and NCL penalty term. Specifically, the proposed method equips each designated graph convolutional layer with an MLP that predicts links using extracted features. The prediction results of the same convolutional layer of the sublearner are integrated to calculate the penalty term.

For the implementation and parameter settings of the comparative methods, most of them are inherited from the original papers of each method. For Katz, the damping factor β is set as 0.001. For PR, the damping factor d is set as 0.85. In the Node2Vector (N2V), each node is embedded as a 128-D vector. For the implementation of Deepwalk, the work in [42] is used and embedding is generated using the default settings.

For processing of data, 15% existing links of the graph are removed and taken as positive testing data. The same number of additional nonexistent links is sampled as negative testing data. The remaining 85% existing links as well as the same number negative samples of nonexistent links are utilized to construct the training data [34]. Experiments are implemented on PyTorch in Ubuntu 20.04 OS with an Nvidia GeForce 3090 GPU.

C. Evaluation Metric

1) AUC: It is the area under the ROC curve in which true positive rate T is taken as a function of false positive rate F. T, F, and AUC are defined as

$$T = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

$$F = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{16}$$

$$\mathrm{AUC} = \int_0^1 T(F)dF \tag{17}$$

[5]The data are available at *https://github.com/shamosha/KeywordGraphData*
[6]https://www.semanticscholar.org/cord19

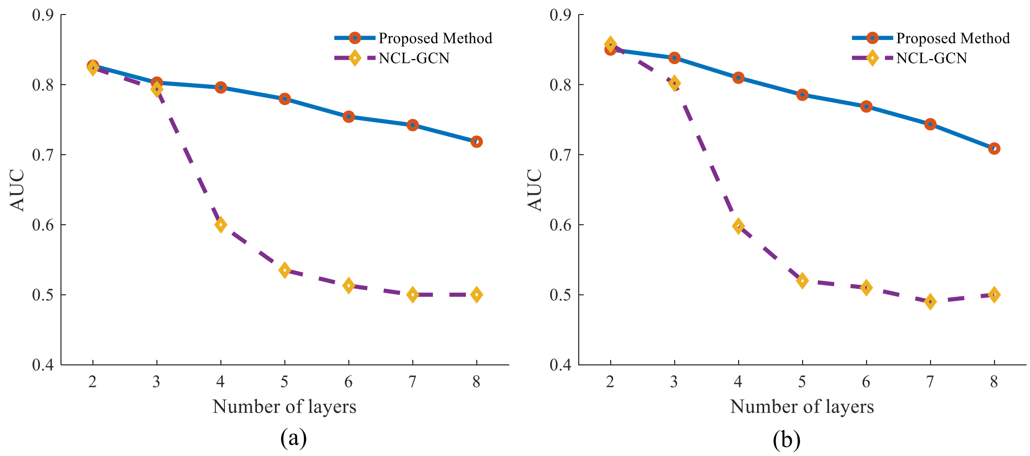


Fig. 3. Performance of the proposed method and the ensemble of GCNs with different numbers of convolution layers. (a) COVID-19 open research dataset. (b) CS dataset.

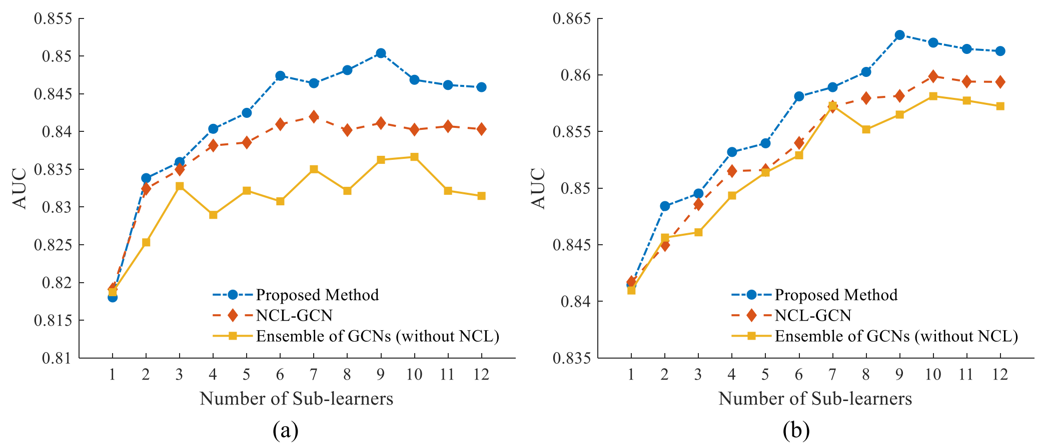


Fig. 4. Performance of three ensemble methods with different numbers of sublearners. (a) COVID-19 open research dataset. (b) CS dataset.

where TP, TN, FP, and FN correspond to the number of true positive, true negative, false positive, and false negative samples, respectively.

2) Average Precision: It is the average precision computed on varying threshold values of recall R. The expression of AP is

$$\mathrm{AP} = \int_0^1 P(R)dR \tag{18}$$

where the definitions of precision P and recall R are expressed as

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{19}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{20}$$

D. Parameter Analysis

The performance of the proposed model depends on four factors: the number of convolutional layers, the number of sublearners, implementation of HNC on different layers, and the weight of HNC separation term. The above hyperparameters are analyzed to show their influence on the performance of the proposed model. Note that if not stated, the number of convolutional layers, the number of sublearners, and the weight of HNC separation term are 2, 3, and 0.5, respectively. Also, the HNC mechanism is implemented in the first convolutional layer.

1) Number of Convolutional Layers: Experiments are conducted on Datasets (A) and (B) to analyze the influence of convolutional layers. The performance of the proposed model and the ensemble of GCNs is shown in Fig. 3. Results show that as the number of convolutional layers increases, the AUC of both the proposed model and the ensemble of GCNs decreases, which may be due to Laplacian oversmoothing.

It is noteworthy that when convolutional layers are more than 4, the performance of NCL-GCNs without adding HNC mechanism declines significantly, but the proposed model is still effective. The HNC promotes the features of low-order subgraphs to be various, thus making the high-order features of different nodes diverse during feature aggregation and performing well.

2) Number of Sublearners: The performance of the proposed model with different numbers of sublearners is given in Fig. 4. It shows that as the number of sublearners increases, the AUC of the proposed model rapidly increases and afterward

TABLE I
PERFORMANCE OF THE PROPOSED MODEL WITH DIFFERENT TRAINING METHODS

Investigated Parameters		AUC GCN	AUC HNC	ACC GCN	ACC HNC	AP GCN	AP HNC	Investigated Parameters		AUC GCN	AUC HNC	ACC GCN	ACC HNC	AP GCN	AP HNC
Batch size	32	85.13	85.41	76.67	78.07	86.63	87.60	Dropout	0	84.92	85.91	75.92	78.07	86.01	87.77
	64	85.23	85.40	76.87	77.96	86.70	87.61		0.1	85.69	86.18	77.01	78.36	86.85	88.13
	128	84.78	85.47	75.91	77.90	86.22	87.59		0.3	85.70	85.88	77.14	78.08	86.88	87.97
	256	84.97	85.32	76.07	77.90	86.43	87.49		0.5	85.07	85.46	75.87	77.97	86.51	87.66
	512	84.87	85.39	76.33	77.85	86.38	87.54		0.7	83.80	84.93	75.78	77.19	85.29	86.35
Learning rate	0.0001	72.95	73.42	63.93	66.09	75.42	76.09	Number of epochs	100	73.90	76.05	65.93	67.63	75.78	77.85
	0.0005	81.25	81.57	72.56	73.29	83.09	83.40		300	84.01	85.23	75.58	76.79	85.51	86.51
	0.001	83.54	84.07	75.00	75.87	85.37	85.68		500	85.06	86.18	76.25	77.84	86.50	87.32
	0.005	85.30	85.48	76.73	78.00	86.70	87.61		700	85.65	86.42	77.13	78.77	86.90	87.58
	0.01	79.05	83.96	69.92	75.78	81.09	85.57		1000	86.48	86.58	77.99	79.08	87.80	88.73
Weight decay	0.0001	85.13	85.37	76.41	77.89	86.59	87.57	Number of sub-learners	1	85.42	85.87	76.70	77.58	86.86	87.19
	0.0002	83.73	85.29	75.27	77.07	85.36	86.58		3	85.69	85.36	77.40	77.94	86.98	87.56
	0.0003	83.39	84.99	74.98	76.54	84.87	86.27		5	86.03	86.06	77.43	77.97	87.25	87.63
	0.0004	82.54	84.30	73.88	76.08	84.19	85.66		7	86.05	86.16	77.47	78.15	87.27	87.73
	0.0005	81.49	83.75	72.17	75.67	83.25	85.14		9	86.04	86.29	77.70	78.03	87.24	87.86

TABLE II
COMPARISON OF THE PROPOSED METHOD WITH OTHER METHODS

	COVID-19 AUC	COVID-19 AP	CS AUC	CS AP	PPI AUC	PPI AP	Average Rank
CN	69.97±0.45	69.89±0.49	77.25±0.36	77.33±0.35	80.55±0.23	80.54±0.28	9.25
JA	68.33±0.75	63.76±1.42	73.30±0.31	67.81±0.81	80.54±0.42	80.55±0.50	10.17
PA	66.79±0.39	75.34±0.54	77.67±0.47	82.84±0.51	59.79±0.63	64.76±0.65	9.50
AA	70.03±0.28	71.15±0.56	77.78±0.34	79.11±0.48	80.54±0.30	80.55±0.35	8.17
RA	69.94±0.34	70.31±0.31	77.94±0.27	79.28±0.27	80.55±0.30	80.57±0.25	7.75
Katz	70.44±0.75	77.78±0.98	77.33±0.38	82.24±0.46	85.63±0.77	86.31±0.33	5.83
PR	83.17±0.60	82.84±0.90	83.67±0.65	82.99±1.40	63.70±0.49	62.53±0.70	7.00
N2V	60.80±0.71	62.52±1.05	65.09±0.54	67.59±0.50	80.93±0.24	85.01±0.16	9.83
Deepwalk	46.42±1.84	46.57±0.77	52.43±0.59	49.04±0.29	61.52±0.30	68.93±0.17	12.50
GCN	83.62±0.59	83.60±0.66	85.81±0.35	87.55±0.41	84.37±0.97	83.64±1.02	3.83
NCL-GCN	84.11±0.89	84.33±1.14	85.99±0.42	88.00±0.47	86.26±1.76	85.59±1.28	2.50
WLMN	79.79±0.32	81.13±0.42	85.76±0.28	87.30±0.41	86.79±0.38	86.80±0.45	3.66
Proposed Method	**85.04**±0.62	**85.09**±0.84	**86.35**±0.48	**88.02**±0.51	**87.10**±0.36	**87.42**±0.66	**1.00**

TABLE III
COMPARISON OF THE PROPOSED METHOD WITH OTHER ENSEMBLE METHODS

	Base learner (GCN)	Bagging			Adaboost			Proposed Method		
Number of sub-learners	-	20	60	100	20	60	100	3	6	9
AUC	89.42	92.64	93.56	93.83	91.43	92.76	93.15	92.98	93.75	**94.05**
ACC	82.86	84.98	86.89	87.56	82.39	85.10	86.27	86.53	87.82	**88.31**
AP	88.28	93.69	94.10	94.39	91.88	93.36	93.83	93.72	94.60	**94.82**
CPU Time (/s)	3.42	48.58	142.32	239.70	58.47	173.00	288.02	11.42	22.09	**34.02**

* CPU Time represents the time spent by the machine to run corresponding method.

tends to remain stable. In addition, as the number of sublearners increases, the performance of the proposed model grows faster and improves greater than ensemble of GCNs, which is the result of explicitly encouraging more diversity.

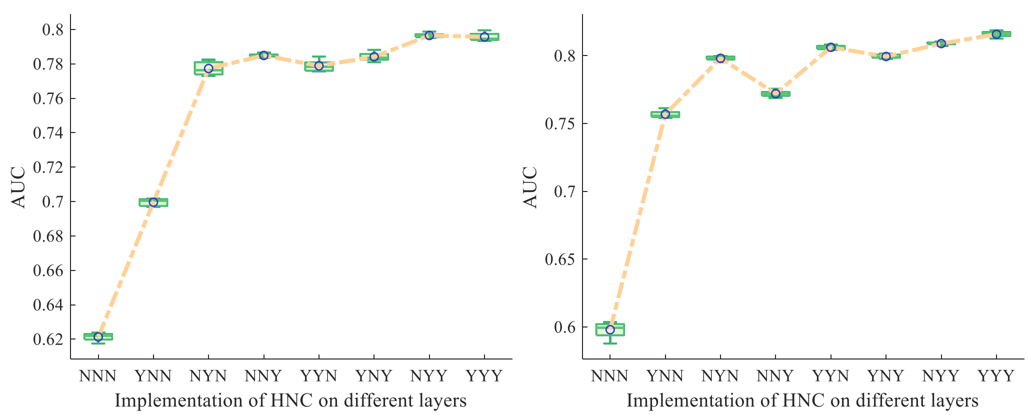
3) Implementation of HNC on Different Layers: To analyze the effectiveness of the HNC, experiments are performed on the proposed model and NCL-GCN, which only considers the negative correlation rather than HNC (introduced in Section III-B), with different numbers of sublearners. The result is shown in Fig. 4, which indicates that the proposed model performs better than NCL-GCN in all observed cases.

To show the influence of different implementations of HNC, experiments are conducted on the proposed method with different H_{ie}'s. The experiments use four-layer GCN as sublearners and the layers applied with HNC are changed to demonstrate the trend of performance. The result is shown in Fig. 5. It shows that the best performance is achieved when the last two layers or all layers are applied with HNC, whereas the worst performance occurs when HNC is only applied in the first layer (except for the non-HNC case). According to the results, it is more effective to improve the performance of the model when the negative correlation constraint is imposed on the multiple high layers simultaneously.

4) Weight of HNC Separation Term: To indicate the influence of the weight of HNC separation term on performance, experiments are conducted with different weights on Datasets (A) and (B). Note that HNC is applied on the last convolutional layers. The result shown in Fig. 6 indicates that the performance of the proposed model tends to increase first and decrease afterward as the weight of HNC separation term increases.

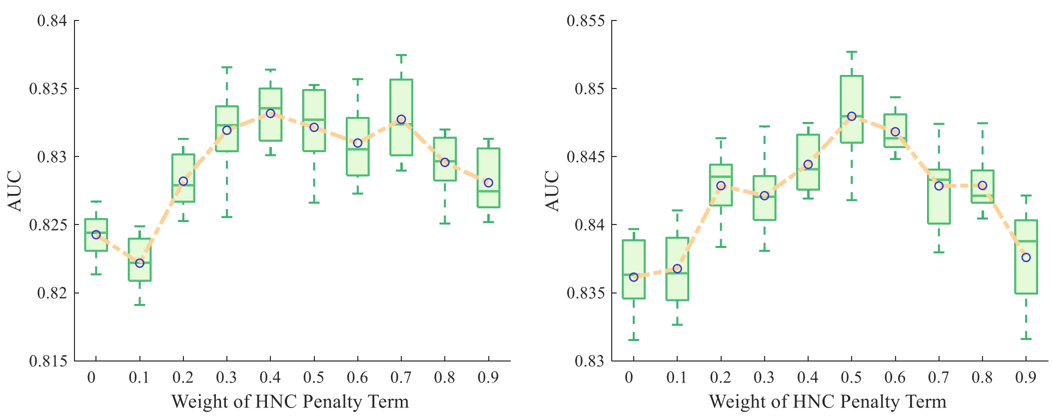
TABLE IV
TYPICAL EXAMPLES OF EFFECTIVE NEW RESEARCH IDEAS

Domain	Keyword 1	Keyword 2	Research Publication	Published Date
COVID-19	Chloroquine	Hydroxychloroquine	Saqrane *et al.* [45]	2020/5
	Generalized additive model	COVID-19	Ma *et al.* [46]	2020/7/1
	Coronavirus	Angiotensin-converting enzyme 2	Batlle *et al.* [48]	2020/3/13
	NK cells	Sindbis virus	Baxter *et al.* [49]	2020/1/16
	Ebola	Stigma	James *et al.* [50]	2020/2/5
	Coumarins	Antiviral activity	Gupta *et al.* [51]	2020/8/8
	Corticosteroids	2019-ncov	Valencia *et al.* [52]	2020/5/24
	Paramyxovirus	Entry inhibitor	Aggarwal *et al.* [53]	2020/3/20
	Alphavirus	Diagnostics	Akhrymuk *et al.* [54]	2020/1/28
	Transcriptomic	Meta-analysis	Barral-Arca *et al.* [55]	2020/3/6
CS	Music generation	Self-attention	Zhang *et al.* [47]	2020/7/2
	Pose estimation	Lie groups	Joukov *et al.* [56]	2020/3
	Quadratic programming	Graph convolutional network	Zhou *et al.* [57]	2020/4
	Finite-time consensus	Switching signals	Zou *et al.* [58]	2020/5
	Multi-view clustering	Augmented Lagrangian multiplier	Zhang *et al.* [59]	2020/1/1
	Hesitant fuzzy information	Mathematical programming	Ranjbar *et al.* [60]	2020/2
	Generative adversarial networks	Visual tracking	Du *et al.* [61]	2020/4/7
	Neurocontrollers	Parsimonious model	Gan *et al.* [62]	2020/2



(a) (b)

Fig. 5. Performance of the proposed method with different HNC implementations. (a) COVID-19 open research dataset. (b) CS dataset. *The experiment uses four-layer GCN, and the first three layers may be applied with HNC. "Y"/"N" indicates whether to implement HNC on the corresponding layer. For example, "YNN" denotes that only the first convolutional layer is applied with HNC. Note that "NNN" is the case that no layer is applied with HNC, and the proposed method becomes NCL-GCN in this case.



(a) (a)

Fig. 6. Performance of the proposed model with different weights of penalty terms. (a) Dataset (A). (b) Dataset (B).

E. Robustness Analysis

To demonstrate the robustness of the proposed model to the training method, experiments are carried out on different training parameter settings. The result is reported in Table I. Specifically, six groups of experiments were conducted in this section. Each group of experiment focuses on a specific training parameter and studies the effect of changing the investigated parameter on the performance of the proposed method. It could be found that the proposed method is effective

TABLE V
RESEARCH IDEAS WHICH MAY BE PUBLISHED

Domain	Keyword 1	Keyword 2
COVID-19	SARS-CoV-2	Epitope prediction
	Corticosteroids	Neutrophil-to-lymphocyte ratio
	Mesenchymal stem cells	COVID-19
	Antibodies	Hemagglutination
	Ambient temperature	Sensitivity analysis
CS	Swarm algorithms	Parameter adaptation
	Consensus tracking problem	Event-triggered control strategy
	Stance detection	Hierarchical attention
	Convex combination technique	Fuzzy min–max neural network
	Side information	Cold start problem

TABLE VI
PREDICTED RESULTS FOR KEYWORD LINKS RELATED TO THIS ARTICLE

Keyword 1	Keyword 2	Result
Convolution neural network	Negative correlation	predicted
Ensemble	Citation networks	predicted
Keyword graph	Link prediction	-
Link prediction	Ensemble model	predicted
Negative correlation	Ensemble	predicted
Negative correlation	Link prediction	predicted
Negative correlation learning	Research idea discovery	-
Research idea discovery	Keyword graph	-

in the selection interval of a wide range of training parameters. Table I also shows the prediction results of the GCN with the same parameter settings, which is the prototype of single sublearners. Through the comparison of the two methods, it can be seen that the proposed method is always better than GCN with the same training parameters in different situations. This phenomenon indicates that the proposed method, as an ensemble model, outperforms a single sublearner. Therefore, the robustness of the proposed method could also be guaranteed by GCN.

V. DISCUSSION

A. Comparison With Other Link Prediction Models

To verify the performance of the proposed model, this article employs 12 comparative methods to conduct experiments, including low-order heuristic method: CN, JA, PA, AA, and RA; high-order heuristics method: Katz and PR; random walk-based method: Deepwalk and N2V; and 4) neural network-based method: GCN, NCL-GCN, and WLMN. The experiments are conducted on Datasets (A), (B), and (C) and the result could be referred to in Table II.

The AUC and AP for the different models indicate that the performance of the proposed model is better than other baseline methods. Specifically, the result of Friedman test [43] based on the performance ranking illustrates that there is statistically a significant difference in the effectiveness of the 13 methods, $\chi^2(12, n = 6) = 55.58$ and $p < 0.01$. In addition, the Wilcoxon signed-rank *post hoc* test [44] is performed on the results of multiple experiments, which verifies that there are significant improvements in performance over the NCL-GCN ($p < 0.05$) and WLMN ($p < 0.05$). Besides, the poor results of traditional heuristic methods indicate that these methods have difficulty in handing datasets with diverse link patterns, which illustrates the effectiveness of learning different aspects of data.

B. Comparison With Other Ensemble Methods

The proposed method is compared with other ensemble methods, as shown in Table III. The utilized graph is constructed by randomly selecting nodes and edges from Dataset (A), containing 200 nodes and 1275 edges. AUC, accuracy (ACC), AP, and time for training the model are chosen to demonstrate the performance of models. Experiments indicate that the proposed method surpasses other ensemble methods. In particular, the proposed method could achieve the effect of other integrated methods with hundreds of sublearners by training a small number of sublearners, which also consumes less time.

C. Discovery of Research Ideas

To demonstrate the effectiveness of the proposed model to discover new research ideas, partial prediction results are shown in Tables IV and V. Table IV presents the representative samples of effective results and Table V shows the keywords that may produce new research ideas that have not been published yet.

The prediction result indicates that the proposed model could discover potential research ideas in different domains. For example, keywords in the medical domain "chloroquine" and "hydroxychloroquine" are predicted to co-occur, and they are used in COVID-19 disease treatment [45]. Besides, the keyword "generalized additive model (GAM)" in the statistics domain is predicted to deal with issues of "COVID-19" in the medical domain. In reality, GAM is utilized to explore the amount of COVID-19 confirmed cases at different temperatures [46]. In the CS domain, the keyword "self-attention" is predicted to applied in the task of "music generation," which corresponds to the fact that Zhang [47] utilized self-attention to impose an explicit musicality regularization in the task of music generation.

Besides, an interesting phenomenon was found, i.e., the model proposed in this article could discover the keyword links related to the research idea of this article. The corresponding prediction results are shown in Table VI. It can be seen that the link of "negative correlation" and "link prediction" is predicted, which is directly relevant to this article. The links of "link prediction" and "ensemble model" are predicted, which is partially related to this article. This phenomenon hints at the effectiveness of the proposed method. However, it has to be pointed out that the links related to "keyword graph" and "research idea discovery" are not successfully predicted, which are also the keywords of this article. This is because of the absence of this keyword in the graph. If a word is not used as a keyword by other papers, it is not included in the graph and related links may not be predicted. Therefore, the method is more inclined to predict the relevance between existing research concepts. It is impossible to predict the connections when the nodes are unknown or generate new nodes. Achieving this may require models to be more creative, which is a challenge for existing ML-based methods.

D. Future Work

Although the proposed model achieves its objectives, there exist a number of issues which worth future research. The granularity of keywords may lead to an ineffective prediction of research ideas. For example, the keyword "antibodies" in Table V is a coarse-grained keyword that may not help in practical research. It is necessary to further consider this issue. In addition, the results could be improved using knowledge. For example, more accurate predictions may be obtained using inference methods based on knowledge graphs. Furthermore, the prediction of domain-specific research ideas may be more popular among researchers. Technologies, such as keyword classification and community division, may be effective for that purpose.

VI. Conclusion

This article explores a method for discovering new research ideas from the perspective of link prediction. An ensemble method of link prediction is proposed to predict the possibility of keyword pairs appearing in future publications. To improve the performance and time consumption for the ensemble model, an HNC mechanism is designed. This mechanism applies negative correlation in the different hierarchical features of the sublearners, taking less time and enhancing the ability to learn features from multiple aspects and different order subgraphs. Moreover, this article discusses the characteristic of HNC theoretically. Experiments are implemented to analyze the impact of hyperparameters on model performance. Meanwhile, the effectiveness of the proposed method for discovering new research ideas in domains related to COVID-19 and CS is discussed. Experimental results indicate that the proposed model outperforms the state-of-the-art models in predicting research ideas.

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Lyuzhou Chen received the B.Sc. degree from the University of Science and Technology of China, Hefei, China, in 2020, where he is currently pursuing the Ph.D. degree with the School of Data Science.

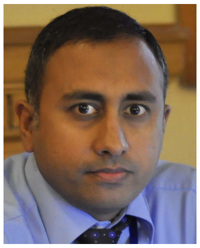
His current research interests include ensemble learning, knowledge engineering, and causal learning.

Xiangyu Wang received the B.Sc. degree from Donghua University, Shanghai, China, in 2015. He is currently pursuing the Ph.D. degree with the School of Data Science, University of Science and Technology of China, Hefei, China.

His research interests include knowledge engineering and machine learning.

Taiyu Ban received the B.Sc. degree in computer science and technology from the School of the Gifted Young, University of Science and Technology of China, Hefei, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, University of Science and Technology of China.

His current research interests include machine learning and knowledge engineering.



Muhammad Usman received the B.Sc. degree from International Islamic University, Islamabad, Pakistan, in 2006, and the master's degree from the Shaheed Zulfikar Ali Bhutto Institute of Science and Technology (SZABIST), Islamabad, in 2017. He is currently pursuing the Ph.D. degree in computer science with the School of Computer Science and Technology, University of Science and Technology of China (USTC), Hefei, China.

Since 2010, he has been working at the Pakistan Scientific and Technological Information Center, MoST, Islamabad, as a Database Administrator. He has published his work at international conferences, journals, and books. His recent research interests include data stream mining, machine learning, data mining, data warehousing, and knowledge discovery.

Shikang Liu received the B.Sc. degree from the University of Science and Technology of China, Hefei, China, in 2021, where he is currently pursuing the M.Sc. degree with the School of Computer Science and Technology.

His current research interests include time-series classification and reservoir computing.

Derui Lyu received the B.Sc. degree in intelligence science and technology from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2021. He is currently pursuing the M.Sc. degree with the School of Computer Science and Technology, University of Science and Technology of China, Hefei, China.

His current research interests include machine learning and knowledge engineering.

Huanhuan Chen (Senior Member, IEEE) received the B.Sc. degree from the University of Science and Technology of China (USTC), Hefei, China, in 2004, and the Ph.D. degree in computer science from the University of Birmingham, Birmingham, U.K., in 2008.

He is currently a Full Professor with the School of Computer Science and Technology, USTC. His current research interests include neural networks, Bayesian inference, and evolutionary computation.

Dr. Chen received the 2015 International Neural Network Society Young Investigator Award, the 2012 IEEE Computational Intelligence Society Outstanding Ph.D. Dissertation Award, the IEEE TRANSACTIONS ON NEURAL NETWORKS Outstanding Paper Award (bestowed in 2011 and only one paper in 2009), and the 2009 British Computer Society Distinguished Dissertations Award. He is also an Associate Editor of the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS and the IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTATIONAL INTELLIGENCE.